Quantum and classical physics-informed networks for laser-pulse thermoelasticity: nominal and output-affecting parameter comparisons

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Abstract

Laser-pulse heating couples thermal diffusion to elastic-wave propagation, providing a demanding test of physics-informed learning. We compare quantum and classical networks against corrected exact thermoelastic fields, using internally prespecified research questions and protocol, with execution changes disclosed. Under the executed recipe, the classical multilayer perceptron has lower mean temperature and stress errors in every nominal-count and output-affecting-count comparison (Tables 6 and 7). The larger circuit takes 171.7–228.3 times the nominally matched network's recorded CPU runtime (Table 6). The full physics objective raises one selected gradient variance, but finite-range and tail slopes establish neither asymptotic decay nor barren-plateau mitigation (Table 8; Supplementary Table S9). Across the tested parameter and collocation budgets and the production training-label sweep conditional on separately tuned hyperparameters, the circuit shows no stress-efficiency advantage over the multilayer perceptron (Table 9). Supervision lowers the circuit's attainable stress error but leaves a classical accuracy gap (Table 9). Secondary wave speeds preserve the circuit–perceptron ordering (Table 11). Common-grid inference controls show greater stress degradation at the lowest shot budget than under either tested exact-expectation depolarizing channel (Table 10); trained readout gains amplify sampling fluctuations. The shared initial-rate exclusion leaves part of the continuum stress evolution to model-dependent implicit selection. Together with limited replication of the efficiency sweeps and contention-affected timings, this restricts the comparison to the declared recipe. The study establishes no quantum advantage.

Highlights

- Nominal and output-affecting count comparisons favor the classical MLP.
- No tested production-budget stress comparison favors the circuit over the MLP.
- Finite-range gradient slopes do not establish asymptotic decay.
- Trained readout gains amplify finite-shot inference errors.
- Initial-rate exclusions leave part of the stress evolution unconstrained.

Keywords: physics-informed neural networks; quantum machine learning; laser heating; thermoelasticity; gradient variance; readout noise.

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1 Introduction

Laser absorption produces localized heating, constrained thermal expansion, and propagating stress. Predicting these coupled fields supports pulse selection for laser processing, thermal-stress control, and interpretation of thermal barrier coating tests. The semi-infinite pulse-heating model of Al-Qahtani (2026) offers an exact-reference benchmark for this purpose. Its simplicity is deliberate: a homogeneous half-space cannot predict coating-interface failure, plasticity, or material removal. More elaborate thermoelastic descriptions include non-Fourier transport and strain feedback (Wang and Xu, 2001); the present benchmark isolates Fourier conduction and one-way thermal forcing of a stress wave.

Physics-informed neural networks (PINNs) approximate fields while penalizing equation and boundary residuals (Raissi et al., 2019). Thermal-mechanical learning for laser metal deposition provides an application precedent (Sharma and Guo, 2024). Accurate stresses, rather than an aggregate training loss alone, matter in such applications. Exact fields permit separate assessment of temperature, stress, and computational expense without uncertainty from a learned reference.

Differentiable quantum circuits can solve differential equations (Kyriienko et al., 2021); repeated data encoding enlarges their function representation (Pérez-Salinas et al., 2020). Quantum and hybrid PINNs have been compared with classical networks (Trahan et al., 2024). In particular, Berger et al. (2025) report favorable comparisons using trainable classical embeddings and equal parameter counts. Heat-equation and parabolic-PDE studies (Binandeh Dehaghani et al., 2025; Tran et al., 2026b), adaptive loss weighting (Setty et al., 2025), and a recent embedding comparison (Tran et al., 2026a) establish substantial prior work. A quantum physics-informed graph-network proposal for laser cutting also exists (Ruhi et al., 2023). The contribution here is a controlled thermoelastic benchmark, not a new general quantum-PINN method or a broad priority claim for laser heating.

Trainability and prediction accuracy are separate questions. Barren-plateau results depend on circuit ensembles, depth, cost locality, and noise (McClean et al., 2018; Cerezo et al., 2021; Wang et al., 2021a). A PDE-based mitigation proposal uses qubit-indexed finite-difference field values (Ukwatta Hewage et al., 2026); our coordinate-encoded circuit instead differentiates fixed observable readouts. Consequently, its gradient diagnostic does not directly test that proposal's representation or locality argument. Increasing a loss's scale also increases gradient variance without establishing improved trainability.

Classical controls are essential to a meaningful quantum comparison (Bowles et al., 2024). Encoding-dependent Fourier representations (Schuld et al., 2021), classical surrogates of trained quantum models (Schreiber et al., 2023), and sufficient conditions for random-feature dequantization (Sahebi et al., 2025) motivate a spectral comparator. Here it is an independently trained classical spectral baseline motivated by dequantization results, with no claim to reproduce the circuit's learned frequencies or function.

The internal brief asks:

1. **RQ1:** Does a quantum PINN match a classical PINN with the same parameter count on these fields?

2. **RQ2:** Does the physics PDE-residual loss change barren-plateau behavior, measured through gradient variance versus qubit count and depth, compared with a data loss?

3. **RQ3:** Is the quantum PINN more parameter-efficient or data-efficient?

The brief supplies no directional statistical hypotheses, equivalence margin, or decision rule. We therefore answer descriptively, without retrospective confirmatory hypotheses. For RQ2, the exe-

cuted statistic uses the full physics objective, including boundary and initial-rate terms. For RQ3, the label comparison concerns production training labels conditional on separately selected hyperparameters, not total exact-label acquisition (Section 3.3). The study follows internally prespecified research questions and protocol, with the execution changes disclosed in Supplementary Section S1. Its contributions are:

- replicated compact-model comparisons on coupled heat–stress fields, with nominal and output-affecting parameter accounting;
- corrected closed forms checked by inverse-Laplace evaluation and an independent finite-difference formulation;
- characterization of readout-gain amplification in selected trained thermoelastic circuits, using common-grid clean and noisy controls.

The general link between observable coefficients and sampling variance, including variance regularization, has prior precedent (Kreplin and Roth, 2024). The contribution is its measured manifestation here.

2 Problem statement and exact solution

2.1 Physical assumptions and dimensional conversion

Consider a homogeneous isotropic half-space with constant coefficients, Beer–Lambert absorption, an insulated traction-free surface, and uniaxial strain. Let $X, \tau, \Theta, u, \Sigma$ denote dimensional depth, time, temperature rise, displacement, and tensile-positive axial stress. The constitutive law and momentum balance are (Wang and Xu, 2001)

$$\Sigma = M u_X - b_{\mathrm{th}}\Theta, \qquad \rho u_{\tau\tau} = \Sigma_X, \tag{1}$$

$$M = \lambda + 2\mu = \rho c_{\mathrm{el}}^2, \qquad b_{\mathrm{th}} = \frac{E\alpha_T}{1-2\nu}. \tag{2}$$

Here α_T is the linear expansion coefficient. Eliminating u gives

$$\Sigma_{XX} - c_{\mathrm{el}}^{-2}\Sigma_{\tau\tau} = c_2\Theta_{\tau\tau}, \qquad c_2 = \frac{b_{\mathrm{th}}}{c_{\mathrm{el}}^2}, \qquad [c_2] = \mathrm{Pa\,s^2\,m^{-2}\,K^{-1}}. \tag{3}$$

The coefficient $b_{\mathrm{th}}/(\rho c_{\mathrm{el}}^2) = b_{\mathrm{th}}/M$, with units $\mathrm{K^{-1}}$, belongs to the displacement equation multiplying Θ_X; it is not the coefficient in Eq. (3).

With absorption coefficient δ, conductivity k, absorbed intensity I_1, and thermal diffusivity $\alpha_{\mathrm{th}} = k/(\rho c_p)$, define

$$x = \delta X, \quad t = \alpha_{\mathrm{th}}\delta^2\tau, \quad \Theta_* = \frac{I_1}{k\delta}, \quad T = \frac{\Theta}{\Theta_*}, \quad \gamma = \frac{c_{\mathrm{el}}^2}{\alpha_{\mathrm{th}}^2\delta^2}. \tag{4}$$

The original stress scale is $\Sigma_0 = I_1 c_2 c_{\mathrm{el}}^2/(k\delta) = b_{\mathrm{th}}\Theta_*$. The benchmark uses $\sigma = \gamma\Sigma/\Sigma_0$, giving the explicit conversion

$$\boxed{\Sigma\,[\mathrm{Pa}] = \frac{E\alpha_T I_1}{(1-2\nu)\gamma k\delta}\,\sigma.} \tag{5}$$

Dimensional material and pulse scales must be supplied before interpreting a predicted stress in pascals. All fields and coordinates below are dimensionless; runtimes are in seconds.

The benchmark equations and complete physical initial data are

$$T_t - T_{xx} = e^{-x} f(t), \qquad \sigma_{xx} - \gamma^{-1}\sigma_{tt} = T_{tt}, \tag{6}$$

$$T_x(0,t) = 0, \qquad \sigma(0,t) = 0, \tag{7}$$

$$T, \sigma \longrightarrow 0 \quad (x \longrightarrow \infty), \qquad T(x,0) = \sigma(x,0) = 0, \tag{8}$$

$$T_t(x,0^+) = e^{-x} f(0^+), \qquad \sigma_t(x,0^+) = -\gamma e^{-x} f(0^+), \quad x > 0. \tag{9}$$

The stress rate follows from initially zero material velocity and the constitutive law. The heat equation has no strain feedback. Phase change, plasticity, ablation, interfaces, and temperature-dependent properties are excluded.

2.2 Pulses, reference fields, and interfaces

Table 1 defines the pulses and domain. Appendix A gives the corrected closed forms implemented in `ground_truth.py`. Inverse-Laplace evaluation (de Hoog et al., 1982) and an independently discretized evolution problem check these fields; Supplementary Section S7 provides the executed verification records and their limits.

Table 1: Problem and evaluation settings from the executed benchmark. H denotes the Heaviside function; the pulses are causal. Energy is $E_0 = \int_0^\infty f(t)\,\mathrm{d}t$.

Item	Definition or value
Exponential	$f(t) = A_e e^{-\beta t}$, $A_e = 1$, $\beta = 2$, $E_0 = 0.5$
Rectangular (step)	$f(t) = A_s[H(t) - H(t-\Delta)]$, $A_s = 1$, $\Delta = 0.5$, $E_0 = 0.5$
Smooth	$f(t) = A(e^{-\beta_1 t} - e^{-\beta_2 t})$, $A = 4/3$, $\beta_1 = 2$, $\beta_2 = 8$, $E_0 = 0.5$
Main material ratio	$\gamma = 9$; dimensionless wave speed $\sqrt{\gamma} = 3$
Network domain	$x \in [0,6]$, $t \in [0,3]$
Test grid	241 spatial nodes on $[0,6]$ and 240 time nodes on $(0,3]$
E1/E1b replication	Seeds $0,1,2,3,4$ in each cell; 105 E1 and 60 E1b runs
E3	Exponential pulse, $\gamma = 9$; seeds $0,1$; 66 completed runs
Secondary wave speeds	Exponential pulse, $\gamma = 25, 49$; seeds $0,1,2$; 24 runs

Sudden startup launches a derivative discontinuity along $x = \sqrt{\gamma}\,t$. The smooth pulse cancels its leading startup kink. The rectangular pulse also has a temporal interface. At fixed interior x, away from intersection with the pre-existing characteristic, define $[g]_\Delta = g(x,\Delta^+) - g(x,\Delta^-)$; then

$$[T]_\Delta = [\sigma]_\Delta = 0, \qquad [T_t]_\Delta = -A_s e^{-x}, \qquad [\sigma_t]_\Delta = \gamma A_s e^{-x}. \tag{10}$$

These conditions follow from the heat equation and the distributional stress equation; $w = \sigma + \gamma T$ and w_t remain continuous. The interface $t = \Delta$ differs from the delayed surface characteristic $x = \sqrt{\gamma}(t - \Delta)$. The executed objective has no explicit interface penalty; sampled smooth residuals alone do not certify these jump conditions.

2.3 Truncation and supplied information

At the outer boundary in Table 1, exact Dirichlet traces for both fields restrict the half-space solution without an additional artificial-boundary approximation. Every physics-trained model receives these traces. "Physics-only" therefore means no interior training labels; exact boundary

values and separate exact validation labels are still supplied. The fully prescribed reference problem and the relaxed training constraints in Section 3.2 must be distinguished. The data-only E3 sweep varies the production training-label count N_d conditional on separately selected hyperparameters. Its axis excludes exact labels used for tuning and validation diagnostics; Section 3.3 and Table 3 account for that additional information.

3 Models and training

3.1 Circuit and classical comparators

Inputs are $u = x/3 - 1$ and $v = 2t/3 - 1$ for the domain in Table 1. Starting from $|0 \cdots 0\rangle$, each circuit layer applies on each qubit

$$G_{\ell q} = R_Z(\theta_{3,\ell q}) R_Y(\theta_{2,\ell q}) R_Z(\theta_{1,\ell q}) R_X(b_{\ell q} v) R_Y(a_{\ell q} u), \tag{11}$$

with the rightmost gate first, followed by the CZ ring. From the released code, the ring has a single edge for the smallest diagnostic circuit; Table 2 records the convention and initializations. Outputs are

$$y_k = \alpha_k \langle Z_k \rangle + \beta_k, \quad k \in \{0, 1\}, \qquad P_{\text{nom}} = 5 n_q L + 4. \tag{12}$$

Training uses exact statevectors and automatic differentiation through complex arithmetic, including the higher-order graphs needed to differentiate second-order residuals with respect to parameters.

Table 2: Architectures and initialization from the released code and executed coordinate-perturbation audit. P_{out} counts output-affecting coordinates, not proven independent functional dimension. Fixed spectral frequencies are not trainable parameters.

Item	Definition
Circuit	$(n_q, L) = (4, 2), (6, 4)$; $P_{\text{nom}}/P_{\text{out}} = 44/30, 124/88$
Initialization	$a, b \sim \mathcal{N}(0, 1)$; $\theta_j \sim \mathcal{U}(0, 2\pi)$; $\alpha_k = 1$, $\beta_k = 0$
CZ ring	Edges $(q, q+1 \bmod n_q)$ for $n_q > 2$; one edge $(0, 1)$ for $n_q = 2$
MLP	$2 \to h \to h \to 2$, tanh; $P = h^2 + 6h + 2$. E1: $h = 4, 8$, $P = 42, 114$; E1b: $h = 3, 7$, $P = 29, 93$
Spectral	$z = [u, v]B$, $B \in \mathbb{R}^{2\times 8}$, $B_{ij} \sim \mathcal{N}(0, \pi^2)$ fixed; $[\cos z, \sin z] \to h \to 2$, tanh, no phase; $P = 19h + 2$. E1: $h = 2, 6$, $P = 40, 116$; E1b: $h = 2, 5$, $P = 40, 97$
Reference MLP	$2 \to 64 \to 64 \to 64 \to 64 \to 2$, tanh; $P = 12802$
Classical initialization	PyTorch `Linear` defaults after setting the run seed; $P_{\text{out}} = P_{\text{nom}}$ denotes no structurally removed coordinates, not a rank measurement
Matching	Nearest count; prescribed tolerance $\pm 10\%$; all deviations reported, including exceptions
Coordinate audit	Model seed 0; input seed 12345; 64 uniform inputs in $[-1, 1]^2$; additive perturbation $+0.37$ per scalar; inert if maximum output change is below 10^{-12}

The terminal CZ ring commutes with the readouts; final-layer operations on unmeasured qubits and terminal measured-qubit R_Z rotations cannot affect them. Backward causal structure removes further coordinates. The released `functional_count.py` and its executed audit list the inert coordinates underlying Table 2; Supplementary Section S2 extends the accounting to every E3 circuit. These counts do not measure Jacobian rank or rule out dependencies among the remaining coordinates. The perturbation is an angle increment only for rotation coordinates, since encoding scales and affine readout parameters are also tested.

E1 compares registered counts; E1b retunes classical networks toward the circuit's output-affecting counts. The smaller spectral row repeats E1 and is not an output-count match. Its minimum is imposed by the implemented width search, not the architecture; the larger spectral row also slightly exceeds the prescribed tolerance (Table 7). The reference MLP is an attainable-accuracy control. Equal counts do not imply equal capacity, derivative cost, memory, or optimization work.

3.2 Hard initial fields and physics objective

Every family uses

$$T_\vartheta = t y_0(u, v; \vartheta), \qquad \sigma_\vartheta = t y_1(u, v; \vartheta). \tag{13}$$

This enforces initial fields, but leaves their initial rates trainable. Writing $\langle\cdot\rangle_C, \langle\cdot\rangle_B, \langle\cdot\rangle_I$ for sample means over collocation points, shared boundary times, and retained initial-rate locations, the released code uses

$$r_h = T_{\vartheta,t} - T_{\vartheta,xx} - e^{-x} f(t), \qquad r_s = \sigma_{\vartheta,xx} - \gamma^{-1}\sigma_{\vartheta,tt} - T_{\vartheta,tt}, \tag{14}$$

$$\mathcal{L}_h = \langle r_h^2\rangle_C, \qquad \mathcal{L}_s = \langle r_s^2\rangle_C, \tag{15}$$

$$\begin{aligned}\mathcal{L}_B = {} & \langle T_{\vartheta,x}(0,t)^2\rangle_B + \langle \sigma_\vartheta(0,t)^2\rangle_B \\ & + \langle [T_\vartheta(6,t) - T^*(6,t)]^2\rangle_B + \langle [\sigma_\vartheta(6,t) - \sigma^*(6,t)]^2\rangle_B,\end{aligned} \tag{16}$$

$$\mathcal{L}_I = \langle [y_0(x,0) - e^{-x} f(0^+)]^2\rangle_I + \left\langle \left[\frac{y_1(x,0) + \gamma e^{-x} f(0^+)}{\gamma} \right]^2 \right\rangle_I, \tag{17}$$

$$\mathcal{L}_P = \mathcal{L}_h + \mathcal{L}_s + \mathcal{L}_B + \lambda_I \mathcal{L}_I. \tag{18}$$

In the rate terms, $y_k(x, 0)$ abbreviates evaluation at the corresponding scaled coordinates. A star denotes an exact field. All weights and reductions are specified in Table 3; no adaptive weighting is used. Adaptive weighting has been studied in quantum differential equation solvers (Setty et al., 2025) and is a distinct future comparison.

The stress equation requires an initial rate. For nonzero-start pulses, its interior right limit conflicts with the differentiated free-surface condition at the initial corner. Excluding an interval, however, is stronger than excluding that point. The executed cutoff in Table 3 omits 45.12% of the squared nonzero initial-rate profile on the truncated domain, since

$$\frac{\int_0^{x_{\min}} e^{-2x}\,\mathrm{d}x}{\int_0^6 e^{-2x}\,\mathrm{d}x} = \frac{1 - e^{-2x_{\min}}}{1 - e^{-12}}. \tag{19}$$

This is a squared-profile fraction, not mechanical energy.

The retained continuum constraints do not uniquely determine stress. Choose a smooth nonzero g supported inside $(0, x_{\min})$ and add a homogeneous wave v satisfying

$$v_{tt} = \gamma v_{xx}, \quad v(x,0) = 0, \quad v_t(x,0) = g(x), \quad v(0,t) = v(6,t) = 0. \tag{20}$$

It preserves the PDE, stress boundaries, zero initial stress, and every retained rate condition. This function-space argument does not assert multiple exact zero-loss realizations in a finite neural architecture. It establishes that part of the stress evolution is left to model-dependent implicit selection. The experiment compares a common declared recipe against the complete exact reference; it does not certify solution of the fully prescribed initial-boundary-value problem.

Table 3: Sampling, training, and validation settings from the execution record and released code for the reference physics recipe. E3 changes point budgets and, for data-only training, the objective and optimizer schedule (Table 9). Exponential-distribution parameters are means. Boundary counts denote shared time samples for all boundary terms.

Item	Executed setting
Adam stage	2000 steps; $N_c = 512$ resampled collocation points per step; 128 boundary times sampled with replacement from 4096 precomputed exact far-boundary values
Collocation mixture	Half uniform on $[0, 6] \times [10^{-3}, 3]$; half with $x \sim$ Exp(mean 1) clipped to $[0, 6]$ and $t \sim$ Exp(mean 0.7) clipped to $[10^{-3}, 3]$
Initial-rate locations	Collocation x values retained for $x \geq 0.3$, then evaluated at $t = 0$
Loss weights	Each PDE and boundary component has weight 1; $\lambda_I = 1$ in all physics runs, including E2
Adam schedule	Cosine decay to 5% of the initial learning rate
L-BFGS stage	Configured limits: 1000 iterations, 1250 evaluations; optimizer step size 1; strong-Wolfe line search; history 50; gradient tolerance 10^{-12}; change tolerance 10^{-14}; fixed 2048 collocation points and 512 shared boundary times
Rate candidates	$\{10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}\}$
Rate selection	1000 Adam steps per candidate, seed 100; 2000 fixed random validation points; score $e_T + e_\sigma$; never the test grid
Selection grouping	Independently for each pulse, family, and configuration
E3 data-only tuning	Rates selected once at $N_d = 256$: 256 exact training samples and 2000 exact validation samples shared by the 5 candidate rates and both families; seed 100
E3 production validation	Diagnostics only; no parameter updates, early stopping, or checkpoint selection

Table 4 completes the single-seed cutoff comparison. Stress sensitivity is nonmonotone and model-dependent: the declared cutoff gives the lowest tested circuit and reference-MLP stress errors, but the highest of the matched MLP's tested stress errors. The circuit and reference MLP therefore do not establish a generally beneficial cutoff. Temperature ordering differs. None of these outcomes is a replicated effect estimate. The separate weight ablation lowers stress error when the rate term is restored, but not uniformly temperature error (Supplementary Table S7). Both ablations change the temperature-rate and stress-rate penalties together.

Table 4: Near-surface initial-rate exclusion ablation, exponential pulse, $\gamma = 9$, seed 0, $N_c = 512$. Every row is an individual-run outcome ($n = 1$), with no standard deviation. Learning rates are 0.1 for VQC/MLP and 0.01 for MLP-ref. The replay verifies the reference-MLP thread setting and is not an extra cutoff replicate.

Model	$P_{\text{nom}}/P_{\text{out}}$	$x_{\min}$	e_T	e_σ
VQC	124/88	0	0.1824	0.9644
VQC	124/88	0.1	0.1709	1.1081
VQC	124/88	0.3	0.1939	0.8785
MLP	114/114	0	0.1330	0.2695
MLP	114/114	0.1	0.1294	0.2393
MLP	114/114	0.3	0.1959	0.3409
MLP-ref	12802/12802	0	0.0176	0.1001
MLP-ref	12802/12802	0.1	0.0143	0.0705
MLP-ref	12802/12802	0.3	0.0085	0.0496
MLP-ref	12802/12802	0.6	0.0166	0.0617
MLP-ref	12802/12802	1	0.0082	0.0732
MLP-ref replay	12802/12802	0.3	0.0074	0.0461

The VQC/MLP cutoff-0.3 rows reuse their E1 seed-0 records. Other cutoff runs use the ablation launcher's 2 threads. The reference cutoff-0.3 run ends at 1060 L-BFGS closures; the 3-thread replay reproduces E1 exactly at 1065 closures. Supplementary Section S1 discusses the comparison. The reference stress-error span across its five 2-thread cutoff runs is 0.0506.

3.3 Training, tuning, and computational accounting

Table 3 gives the common nominal schedule. Initial learning rates are selected independently by minimizing $e_T + e_\sigma$ on fixed random validation points, never the test grid. Complete candidate histories and selected scores are released (Supplementary Table S4). The upper boundary is selected in 11/21 E1 and 10/12 E1b cells; one E1 cell selects the lower boundary (Supplementary Table S4). These are conditional comparisons under the executed tuning and training recipe, not evidence that each architecture attained its best achievable accuracy. Selection after Adam also need not identify the best subsequent Adam–L-BFGS endpoint.

The E3 label sweep measures production training-label efficiency conditional on separately selected hyperparameters. Its data-only rates were selected once at $N_d = 256$, using 256 exact training samples and 2000 exact validation samples shared across the five candidate rates (Table 3; Supplementary Table S4). This fixed tuning overhead is outside the production N_d axis. Production runs also compute exact-label validation diagnostics, but these neither drive updates nor select checkpoints. Thus N_d is not the total exact-label requirement for developing or evaluating the tuned learner.

Actual L-BFGS closure calls are objective evaluations, distinct from optimizer iterations, and vary across runs (Supplementary Table S5). The timer covers training, periodic validation, and final test-grid evaluation, excluding model construction, exact-data preparation, and tuning. E1

uses the shared CPU with the concurrency and thread settings in Supplementary Table S3; other launchers differ. Consequently, the reported durations are implementation-specific shared-machine timings, not isolated throughput, total campaign cost, or quantum-hardware cost. E3 timings are contention-affected observed durations, unsuitable for isolated-throughput comparisons; E1 is retained as the primary cost comparison. A reference-MLP replay at E1's thread setting reproduces its numerical history and endpoint; the different-thread cutoff run diverges only during L-BFGS (Table 4; Supplementary Section S1). Thread-dependent reduction order amplified by the line search is a plausible explanation, not a separately tested causal diagnosis.

3.4 Metrics and diagnostic protocols

For $F \in \{T, \sigma\}$ and test grid $\mathcal{G}$,

$$e_F = \left[\frac{\sum_{z\in\mathcal{G}} |F_\vartheta(z) - F^*(z)|^2}{\sum_{z\in\mathcal{G}} |F^*(z)|^2}\right]^{1/2}, \qquad a_F = \max_{z\in\mathcal{G}} |F_\vartheta(z) - F^*(z)|. \tag{21}$$

E1, E1b, and secondary-speed errors use means and sample standard deviations with denominator $n-1$. E3 instead reports means and both seed values or their observed range, because each cell has only two seeds (Table 9); these ranges are not confidence intervals. All completed runs, including poor fits, are retained. Best seeds are descriptive minima. No significance or equivalence test is used.

E2 compares the full physics objective with

$$\mathcal{L}_D = \langle (T_\vartheta - T^*)^2 \rangle_D + \langle (\sigma_\vartheta - \sigma^*)^2 \rangle_D, \qquad \mathcal{L}_C = \mathcal{L}_P + \mathcal{L}_D. \tag{22}$$

The selected derivative, sampling, and variance denominator are specified in Table 5. Multiplying an objective by a constant multiplies gradient variance by its square; absolute variance increases must therefore be interpreted with the stated normalization.

Table 5: Diagnostic settings from the released code and executed records.

Diagnostic	Setting
E2 circuit scan	$n_q = 2, 3, 4, 5, 6, 7, 8, 9, 10$; $L = 2, 4, 8$; 200 random initializations with seeds 0–199 per cell; exponential pulse, $\gamma = 9$
E2 sampled derivative	First-layer trainable R_Y angle on qubit 0
E2 points	256 fixed uniform interior points, shared between objectives; 64 boundary times; point-set generator seed 0
E2 variance	`numpy.var` with denominator 200 (`ddof=0`)
E2 fit	Ordinary least squares of $\log_{10} \mathrm{Var}(\partial\mathcal{L}/\partial\theta)$ against n_q
E1 gradient summaries	Mean gradient norm over the first and last 50 Adam steps
E4 checkpoints	Lowest-e_σ 2 E1 seeds per pulse/configuration; listed in Supplementary Table S10
E4 evaluation	41×40 grid; binomial samples of each Pauli-Z expectation at $N = 10^3, 10^4, 10^5$ shots per input and observable
E4 channel	Pauli-error depolarization after every layer on every qubit; $p = 0, 10^{-3}, 10^{-2}$; density-matrix inference; exact controls at each p

E4 evaluates selected checkpoints without retraining. Its dispersion is across trained checkpoints, not independent repeated shot trials. The per-qubit channel after each layer is

$$\mathcal{E}_{p,q}(\rho) = (1-p)\rho + \frac{p}{3}(X_q\rho X_q + Y_q\rho Y_q + Z_q\rho Z_q). \tag{23}$$

Here p is total Pauli-error probability. Independent marginal binomial draws estimate the readouts using N measurements per input and observable; commuting readouts could instead share hardware bitstrings. The evaluator uses each checkpoint's run seed for its sampling generator. It does not estimate repeated-shot uncertainty or test noisy optimization.

4 Results

4.1 Nominal and output-affecting parameter comparisons

Table 6 is the numerical anchor for E1; Figure 1 displays its field errors. The nominally matched MLP has lower mean temperature and stress errors in every pulse–configuration cell. The circuit/MLP mean stress-error ratios range from 1.788 to 3.645 (Table 6). These are ratios of means, not paired significance tests. The larger circuit's recorded runtime is 171.7–228.3 times the matched MLP's, compared with 41.7–51.8 for the smaller circuit (Table 6).

Table 6: E1 nominal-count comparisons. Errors and recorded runtime are mean ± sample standard deviation across five seeds. $P_{\text{nom}}/P_{\text{out}}$ lists registered and output-affecting parameter counts, respectively; P_{out} is not a proven independent functional dimension. R_σ in each VQC row is its mean stress error divided by its nominally matched MLP mean. Runtime includes training and both evaluations under the stated concurrent execution conditions.

Pulse	Config.	Model	$P_{\text{nom}}/P_{\text{out}}$	e_T	e_σ	Runtime (s)	R_σ
exp	4q2L	VQC	44/30	0.4699 ± 0.1428	1.1137 ± 0.1711	725.1 ± 68.2	3.645
exp	4q2L	MLP	42/42	0.3178 ± 0.0738	0.3055 ± 0.0608	15.7 ± 1.4	–
exp	4q2L	Spectral	40/40	0.4500 ± 0.1532	0.7826 ± 0.2417	45.0 ± 3.2	–
exp	6q4L	VQC	124/88	0.1779 ± 0.0439	0.7608 ± 0.1884	4188.9 ± 251.0	3.544
exp	6q4L	MLP	114/114	0.0990 ± 0.0584	0.2146 ± 0.1375	19.7 ± 0.5	–
exp	6q4L	Spectral	116/116	0.2401 ± 0.1689	0.5719 ± 0.2035	44.1 ± 3.7	–
exp	–	MLP-ref	12802/12802	0.0125 ± 0.0029	0.0539 ± 0.0076	214.6 ± 9.4	–
step	4q2L	VQC	44/30	0.6688 ± 0.1857	1.1934 ± 0.3074	624.5 ± 15.5	2.506
step	4q2L	MLP	42/42	0.3314 ± 0.0683	0.4762 ± 0.0743	15.0 ± 0.1	–
step	4q2L	Spectral	40/40	0.7012 ± 0.0570	0.9526 ± 0.0595	42.5 ± 3.4	–
step	6q4L	VQC	124/88	0.2579 ± 0.0336	0.8954 ± 0.1136	3644.1 ± 21.1	2.513
step	6q4L	MLP	114/114	0.2105 ± 0.0904	0.3564 ± 0.1015	21.2 ± 2.8	–
step	6q4L	Spectral	116/116	0.3405 ± 0.1200	0.3825 ± 0.0742	47.3 ± 3.4	–
step	–	MLP-ref	12802/12802	0.0221 ± 0.0147	0.0517 ± 0.0086	212.6 ± 7.8	–
smooth	4q2L	VQC	44/30	0.3260 ± 0.0582	0.9826 ± 0.0958	758.4 ± 115.0	1.788
smooth	4q2L	MLP	42/42	0.2220 ± 0.0870	0.5496 ± 0.3482	14.6 ± 1.0	–
smooth	4q2L	Spectral	40/40	0.6403 ± 0.1968	0.9434 ± 0.1900	43.1 ± 8.7	–
smooth	6q4L	VQC	124/88	0.1664 ± 0.0342	0.5733 ± 0.0696	4473.7 ± 466.5	3.364
smooth	6q4L	MLP	114/114	0.0519 ± 0.0143	0.1704 ± 0.0632	19.6 ± 2.6	–
smooth	6q4L	Spectral	116/116	0.2524 ± 0.1288	0.6254 ± 0.2388	43.3 ± 4.1	–
smooth	–	MLP-ref	12802/12802	0.0058 ± 0.0013	0.0296 ± 0.0106	192.8 ± 10.6	–

E1b addresses the inert-coordinate imbalance by comparing the same circuits with retuned smaller classical networks. The MLP again has lower mean errors for both fields in all six cells, with mean stress-error ratios 1.558–4.493 (Table 7). The nominal-count result therefore does not reverse when this imbalance is reduced. This does not establish equal intrinsic model dimension or a universal classical advantage. The spectral baseline is less consistent: its temperature error exceeds the circuit's in five of six E1b cells, and its larger smooth-pulse stress error also exceeds the circuit's (Tables 6 and 7).

Figure 2 shows representative best-stress-seed fields. The reference MLP attains substantially smaller errors under this recipe (Table 6); the images illustrate spatial discrepancies without establishing representational impossibility for any family.

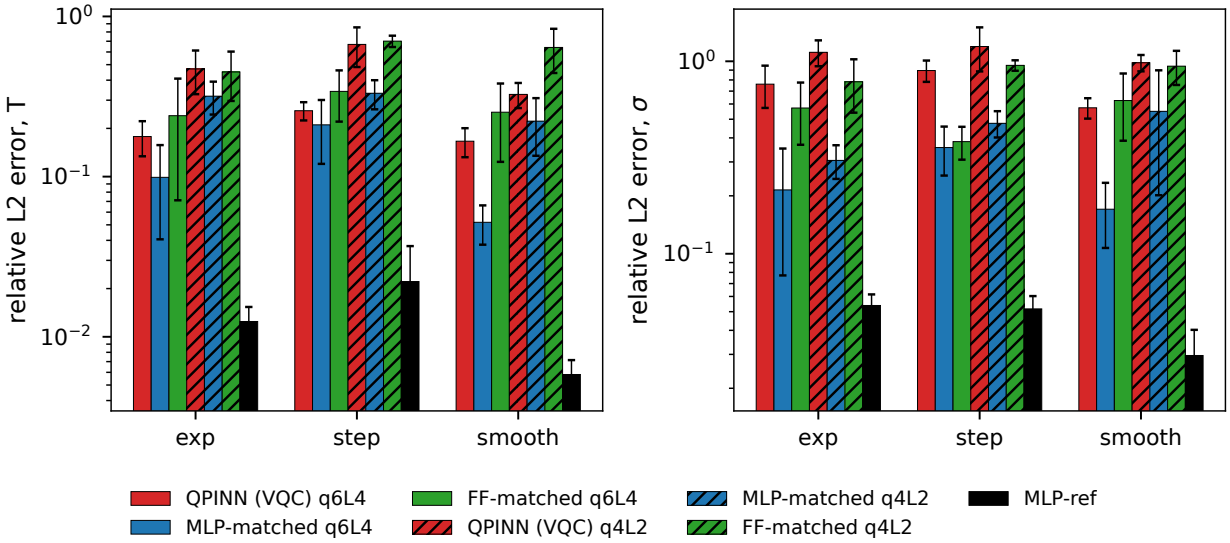


Figure 1: E1 nominal-count comparison at the settings of Tables 1–3. Bars show mean relative temperature and stress errors; error bars are sample standard deviations over five seeds. Solid bars represent the larger configuration and hatched bars the smaller one; the reference MLP is separate. Circuit $P_{\text{nom}}/P_{\text{out}}$ is 124/88 or 44/30, with classical counts listed in Table 6. The plot does not show E1b.

Table 7: E1b classical comparisons targeting circuit output-affecting counts 30 and 88, for circuits with $P_{\text{nom}}/P_{\text{out}} = 44/30$ and 124/88. Each error is mean ± sample standard deviation over five seeds, with learning rates reselected per cell. ΔP is the classical count deviation from the target in percent; R_σ is the corresponding E1 VQC mean stress error divided by the classical mean.

Pulse	Target	Model	$P_{\text{nom}}/P_{\text{out}}$	ΔP (%)	e_T	e_σ	R_σ
exp	4q2L	MLP	29/29	−3.33	0.3322 ± 0.0579	0.4396 ± 0.1554	2.533
exp	4q2L	Spectral†	40/40	+33.33	0.4500 ± 0.1532	0.7826 ± 0.2417	1.423
exp	6q4L	MLP	93/93	+5.68	0.1379 ± 0.0302	0.2857 ± 0.0607	2.663
exp	6q4L	Spectral	97/97	+10.23	0.3162 ± 0.1474	0.6332 ± 0.0497	1.201
step	4q2L	MLP	29/29	−3.33	0.5283 ± 0.1594	0.7658 ± 0.3048	1.558
step	4q2L	Spectral†	40/40	+33.33	0.7012 ± 0.0570	0.9526 ± 0.0595	1.253
step	6q4L	MLP	93/93	+5.68	0.1710 ± 0.0267	0.1993 ± 0.0391	4.493
step	6q4L	Spectral	97/97	+10.23	0.3425 ± 0.1174	0.7213 ± 0.2003	1.241
smooth	4q2L	MLP	29/29	−3.33	0.2652 ± 0.0251	0.4884 ± 0.0163	2.012
smooth	4q2L	Spectral†	40/40	+33.33	0.6403 ± 0.1968	0.9434 ± 0.1900	1.042
smooth	6q4L	MLP	93/93	+5.68	0.0902 ± 0.0258	0.3631 ± 0.2001	1.579
smooth	6q4L	Spectral	97/97	+10.23	0.3688 ± 0.1425	0.8194 ± 0.1841	0.700

† The implemented spectral width search starts at $h = 2$, giving a minimum of 40 parameters. These rows repeat the E1 network outcomes and are not valid output-count matches. A width-one spectral network would have 21 parameters but was not included in the implemented search. The 97-parameter spectral model differs from target 88 by 10.23%, slightly exceeding 10%.

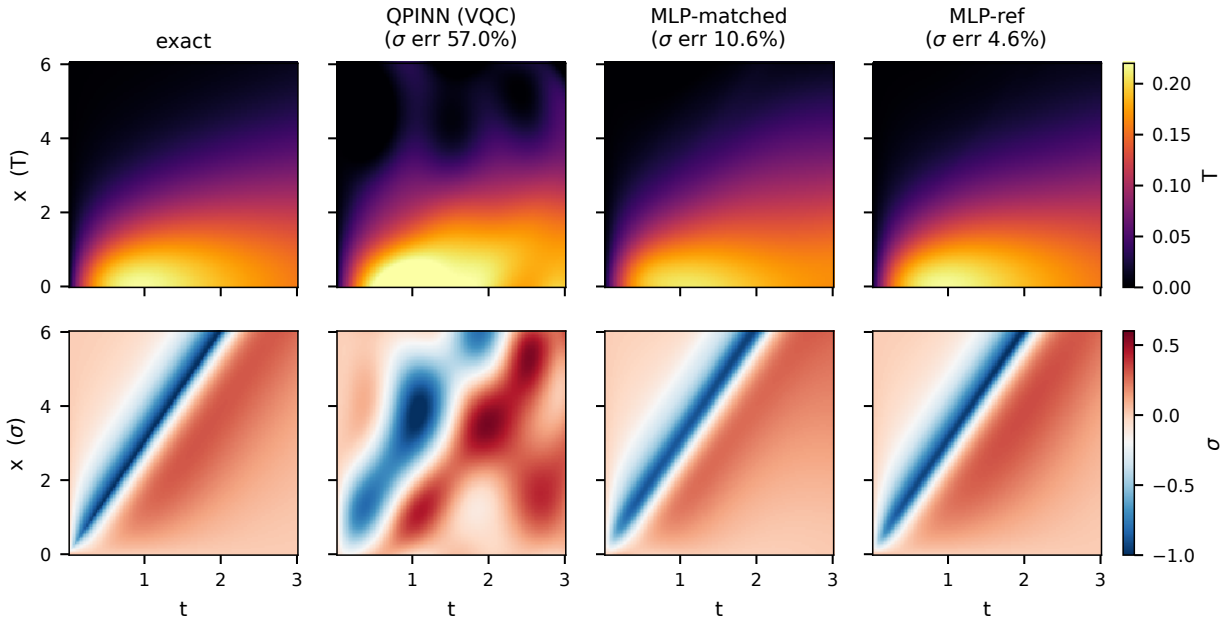


Figure 2: Exact and learned exponential-pulse fields at $\gamma = 9$. Temperature is above stress; columns show the exact solution, the larger circuit ($P_{\text{nom}}/P_{\text{out}} = 124/88$), nominally matched MLP (114/114), and reference MLP (12802/12802). Learned examples use the minimum-stress-error E1 seeds, identified in Supplementary Table S6. Quantitative color bars give dimensionless T and σ. The characteristic $x = 3t$ follows the wave speed in Table 1. These are selected examples, not replicated error summaries.

4.2 Gradient variance

Figure 3 and Table 8 summarize the variance of a selected derivative of the full physics objective. The physics/data variance multipliers range from 5.19–20.31, 70.89–375.41, and 2726.20–9311.63 across the respective depths (Supplementary Table S9). The deepest whole-range slopes are negative, but its largest-qubit refits are positive (Table 8). Thus neither unchanged decay nor persistent exponential scaling is established.

Table 8: Descriptive ordinary-least-squares slopes of $\log_{10} \mathrm{Var}(\partial\mathcal{L}/\partial\theta)$ versus qubit count. The selected derivative is the first-layer R_Y angle on qubit 0; each variance uses 200 initializations. The final row refits the largest-qubit portion of the deepest scan. No slope-equivalence test or asymptotic exponent is inferred.

Depth L	Fitted n_q range	Data	Full physics	Combined
2	2–10	−0.008	+0.027	+0.020
4	2–10	−0.038	−0.076	−0.074
8	2–10	−0.206	−0.227	−0.227
8	7–10	+0.014	+0.020	+0.020

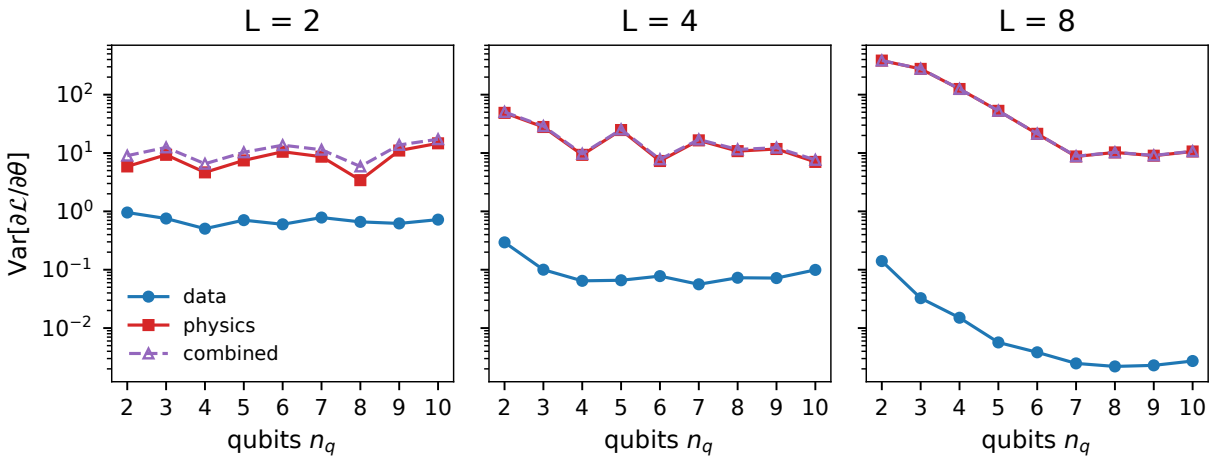


Figure 3: Variance of the first-layer measured-qubit R_Y derivative at random initialization, for the objectives and sampling in Table 5. Integer qubit ticks and distinct markers separate the full physics and combined objectives. The curves use population-denominator variance over 200 initializations per cell; they are not mean–standard-deviation bands. Full values are in Supplementary Table S8 and descriptive slopes in Table 8.

At fixed depth a local readout has a bounded backward causal cone, so indefinite qubit growth cannot be assumed to reproduce a whole-range finite-size fit (Cerezo et al., 2021). This observation limits extrapolation; it does not demonstrate that cone saturation caused the observed tail, whose circuit can still span the tested ring. Individual gradient draws are not retained, and no slope-equivalence test is available. E2 therefore answers a narrower question than a barren-plateau theorem and does not isolate the contribution of the PDE residual from the other physics terms.

4.3 Efficiency

The completed E3 sweeps address parameter count, collocation budget, and production training-label budget conditional on separately selected hyperparameters for the exponential pulse. Table 9 reports means and observed ranges over the declared two seeds; Supplementary Section S9 retains every individual run, matching deviation, learning rate, point budget, runtime, and realized closure count. The parameter and collocation sweeps use the reference physics recipe; data-only training uses $\mathcal{L}_D$, fixed exact interior samples, separately selected rates, and no boundary, rate, or PDE penalty or L-BFGS (Table 9; Supplementary Tables S3–S4).

Parameter count. The nominally matched MLP has lower mean stress error at every tested configuration, by factors 2.0–8.5 (Table 9; Figure 4). Even the best circuit run, $e_\sigma = 0.3951$ at $(n_q, L) = (6, 8)$, exceeds the smallest MLP's mean 0.3067 at 42 parameters. The best circuit mean is 0.4464 at 244/208 nominal/output-affecting parameters (Table 9). Changing the horizontal coordinate to the output-affecting count does not change this accuracy ordering. It does not, however, establish strict Pareto domination of every circuit point: the shallow circuits have fewer output-affecting coordinates than the smallest E3 MLP (Table 9; Supplementary Table S2). E1b supplies the separate comparison with smaller classical networks (Table 7).

The depth-2 circuits with 4, 6, and 8 qubits realize the same 30-coordinate output family, not merely the same output-affecting count (Supplementary Section S2). The final CZ layer cancels in the measured Z expectations; propagation through the preceding ring leaves only qubits $0, 1$ and their neighbors $2, n_q - 1$. Relabeling these active qubits therefore preserves the family. The supplied execution of `family_equivalence.py` confirms the parameter transplant to numerical precision (Supplementary Sections S2 and S8). Their stress means 1.0127, 1.4455, and 2.0390 are differently initialized and trained realizations of that same family with different simulation cost, not a capacity trend (Table 9). At fixed width, increasing depth reduces stress error from 1.4455 through 0.7352 to 0.4464 (Table 9); this remains an observed trend without an identified mechanism. The deepest circuit approaches the MLP's temperature accuracy, 0.0598 versus 0.0557, without an equivalence test. Spectral stress errors fall between circuit and MLP in all but the $(6, 4)$ design, where the spectral mean is slightly larger than the circuit's (Table 9).

The classical rows for circuit targets $(4, 4)$ and $(8, 2)$ share architectures and identical numerical outcomes; their repeated executions are marked in Table 9. Their MLP count deviation is $+10.71\%$; the $(6, 2)$ MLP deviation is -10.94%, both outside the prescribed tolerance (Supplementary Table S2). The $(4, 2)$ and $(6, 4)$ parameter runs exactly reproduce the corresponding E1 numerical histories and errors for the same seeds, although recorded runtimes differ (Supplementary Section S1). They do not add independent replication.

Collocation budget. Increasing the number of residual points does not systematically lower either family's errors or close the stress gap in the tested range (Table 9; Figure 5). Across shared budgets, the circuit/MLP stress-mean ratio is 2.0–4.4, and every completed circuit run has stress error above 0.5 (Table 9). The nonmonotone means and limited seed replication do not prove that either architecture is generally free of collocation limitations.

The circuit uses $N_c = 64, 256, 512, 1024$; the MLP also uses 4096. The baseline at 512 reuses the parameter-sweep runs. The circuit jobs at 4096 were dropped after an initial execution was stopped for cost; the measured-rate projection was about 32400 s per seed after 2100 s of observation, including a fixed L-BFGS set of 16384 points (Table 9; Supplementary Table S3). This is a declared deviation from the planned sweep, not an unreported completed outcome. The retained campaign contains 66 of the planned 68 runs (Supplementary Table S1).

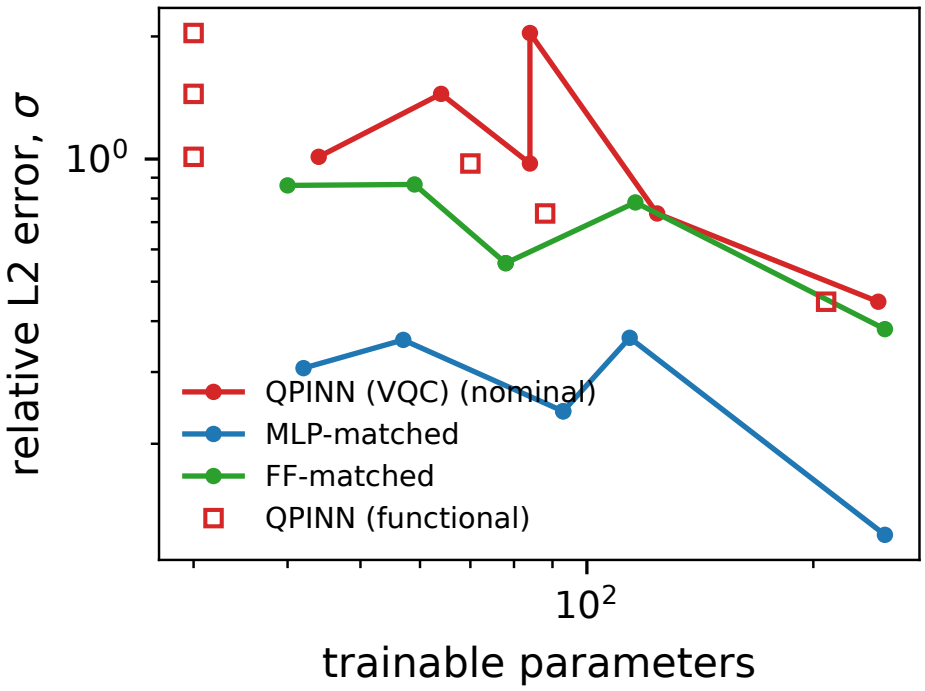


Figure 4: E3 stress error versus parameter count for the exponential pulse, $\gamma = 9$, using means of seeds $0, 1$. Red filled circles use circuit nominal counts; hollow squares show the same errors at output-affecting counts. Blue and green points denote nominally matched MLP and spectral models. Lines connect evaluated architectures and are not fitted Pareto envelopes. The duplicated classical outcomes at the common nominal target are shared. The depth-2 circuits at 4, 6, and 8 qubits are different training realizations of one 30-coordinate output family, as established by the causal structure and supplied parameter-transplant check (Supplementary Sections S2 and S8); their width sequence is not a capacity comparison. Table 9 gives both-seed ranges and Supplementary Table S2 gives every count and matching deviation.

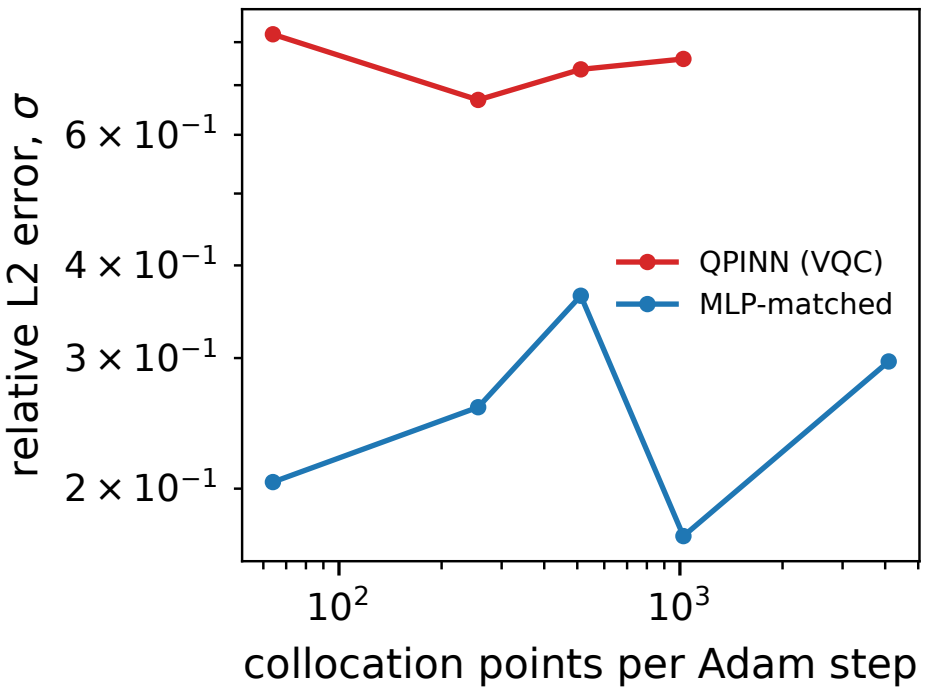


Figure 5: E3 stress error versus collocation points per Adam step. Physics-only training uses fixed L-BFGS sets of $4N_c$ points. Means use seeds $0, 1$ for the $(6, 4)$ circuit (124/88 parameters) and nominally matched MLP (114/114). Circuit budgets are $64, 256, 512, 1024$; the MLP also includes 4096. The circuit's largest planned budget was dropped for cost. The baseline at 512 is reused from the parameter sweep. Observed ranges are in Table 9.

Production training labels. The N_d axis counts exact interior samples used for production updates, conditional on the separate tuning at $N_d = 256$. It excludes the fixed tuning set of 256 exact training samples plus 2000 exact validation samples shared across the five candidate rates, and the production validation diagnostics (Table 3; Supplementary Table S4). Those diagnostics neither drive updates nor select checkpoints. Under supervision, circuit stress means fall from 2.1382 at 16 labels to 0.2918 at 256 and 0.2178 at 1024; the last value is 3.375 times lower than its physics-only mean for the same seeds (Table 9; Figure 6). Thus the same architecture can represent a more accurate stress field than physics training attained. The gap is partly training-objective dependent, rather than evidence of representational impossibility. This comparison changes information, objective, rate, and optimizer schedule, so it cannot establish that the loss landscape is the main cause.

The MLP remains more accurate at every label budget, with stress-mean ratios 1.4–3.3. Its means change little from 0.0884 to 0.0872 between the largest budgets. The circuit at 1024 labels still exceeds the MLP mean at 64 labels, 0.2178 versus 0.1903 (Table 9). These observed production training budgets differ by a factor of 16; they estimate neither an equal-error sample-complexity ratio nor a total label-acquisition ratio. At the smallest budget both families have mean errors above unity for both fields (Table 9).

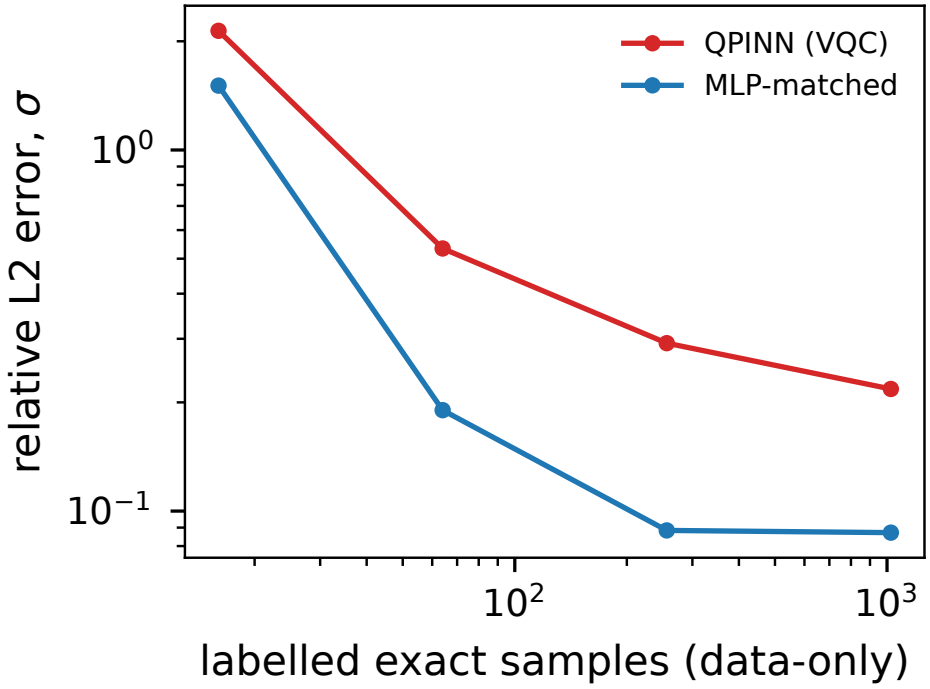


Figure 6: E3 stress error versus fixed exact interior training samples N_d in production runs conditional on separately selected hyperparameters. Means use seeds $0, 1$ for the same circuit and MLP as Figure 5. Training uses 2000 Adam steps, no PDE, boundary, or initial-rate penalty, and no L-BFGS. Validation-selected rates are 0.03 for the circuit and 0.1 for the MLP, selected once at $N_d = 256$ using 256 training and 2000 validation samples shared by the five candidates. The horizontal axis excludes these tuning labels and production validation diagnostics, which neither drive updates nor select checkpoints (Table 3). Table 9 provides the two-seed ranges.

E3 timings are contention-affected observed durations, unsuitable for isolated-throughput comparisons. For example, the deepest physics circuit takes 8683–11552 s, whereas its MLP takes 19–22 s; the supervised circuit ranges from 28 to 156 s across label budgets (Supplementary Table S16). These are not controlled cost ratios. E1 remains the primary wall-clock comparison (Table 6; Supplementary Table S3).

Table 9: E3 errors: mean [minimum, maximum] over seeds $0, 1$ ($n = 2$). P denotes nominal/output-affecting counts. Parameter rows use $N_c = 512$; collocation rows vary N_c with $4N_c$ fixed L-BFGS points; data rows vary N_d conditional on separate tuning (Table 3) and omit all physics losses and L-BFGS. All physics rates are 0.1; data rates are 0.03 (VQC) and 0.1 (MLP). Budgets and per-seed times, closures, and matching deviations are retained in the supplement. The $N_c = 512$ rows reuse the parameter-sweep records. A dagger marks classical outcomes identical to the $(4, 4)$ executions, not additional independent evidence.

Design/budget	Model	P	e_T	e_σ
Parameter sweep: (n_q, L)				
$4, 2$	VQC	44/30	0.4030 [0.2561, 0.5500]	1.0127 [0.8460, 1.1794]
$4, 2$	MLP	42/42	0.2755 [0.2723, 0.2788]	0.3067 [0.2311, 0.3823]
$4, 2$	Spectral	40/40	0.4056 [0.4001, 0.4112]	0.8616 [0.7746, 0.9487]
$4, 4$	VQC	84/70	0.1947 [0.1905, 0.1990]	0.9752 [0.9030, 1.0473]
$4, 4$	MLP	93/93	0.1052 [0.1005, 0.1100]	0.2403 [0.2196, 0.2610]
$4, 4$	Spectral	78/78	0.2830 [0.1796, 0.3864]	0.5552 [0.4204, 0.6899]
$6, 2$	VQC	64/30	0.6330 [0.6162, 0.6498]	1.4455 [1.3497, 1.5413]
$6, 2$	MLP	57/57	0.1530 [0.1020, 0.2040]	0.3598 [0.2854, 0.4341]
$6, 2$	Spectral	59/59	0.3909 [0.2988, 0.4830]	0.8662 [0.8510, 0.8815]
$6, 4$	VQC	124/88	0.1657 [0.1376, 0.1939]	0.7352 [0.5919, 0.8785]
$6, 4$	MLP	114/114	0.1402 [0.0844, 0.1959]	0.3640 [0.3409, 0.3872]
$6, 4$	Spectral	116/116	0.2159 [0.1650, 0.2667]	0.7825 [0.7594, 0.8056]
$6, 8$	VQC	244/208	0.0598 [0.0591, 0.0606]	0.4464 [0.3951, 0.4977]
$6, 8$	MLP	249/249	0.0557 [0.0359, 0.0756]	0.1195 [0.0935, 0.1455]
$6, 8$	Spectral	249/249	0.1420 [0.1327, 0.1513]	0.3823 [0.3583, 0.4064]
$8, 2$	VQC	84/30	0.8757 [0.6694, 1.0819]	2.0390 [1.7132, 2.3649]
$8, 2$	MLP†	93/93	0.1052 [0.1005, 0.1100]	0.2403 [0.2196, 0.2610]
$8, 2$	Spectral†	78/78	0.2830 [0.1796, 0.3864]	0.5552 [0.4204, 0.6899]
Collocation sweep: N_c, $(n_q, L) = (6, 4)$				
64	VQC	124/88	0.2199 [0.1051, 0.3348]	0.8198 [0.8078, 0.8318]
64	MLP	114/114	0.1287 [0.0870, 0.1705]	0.2041 [0.2023, 0.2060]
256	VQC	124/88	0.2158 [0.2140, 0.2176]	0.6687 [0.5376, 0.7998]
256	MLP	114/114	0.1198 [0.0906, 0.1491]	0.2575 [0.2438, 0.2713]
512	VQC	124/88	0.1657 [0.1376, 0.1939]	0.7352 [0.5919, 0.8785]
512	MLP	114/114	0.1402 [0.0844, 0.1959]	0.3640 [0.3409, 0.3872]
1024	VQC	124/88	0.2139 [0.1815, 0.2463]	0.7594 [0.6938, 0.8250]
1024	MLP	114/114	0.0962 [0.0770, 0.1154]	0.1726 [0.0886, 0.2566]
4096	MLP	114/114	0.1165 [0.1118, 0.1213]	0.2968 [0.2496, 0.3441]
Label sweep: N_d, $(n_q, L) = (6, 4)$				
16	VQC	124/88	1.6397 [0.9935, 2.2859]	2.1382 [1.7633, 2.5130]
16	MLP	114/114	1.0870 [0.7236, 1.4504]	1.5074 [1.2370, 1.7778]
64	VQC	124/88	0.2798 [0.2571, 0.3024]	0.5336 [0.4412, 0.6260]
64	MLP	114/114	0.0802 [0.0738, 0.0865]	0.1903 [0.1577, 0.2229]
256	VQC	124/88	0.1907 [0.1873, 0.1940]	0.2918 [0.2882, 0.2953]
256	MLP	114/114	0.1007 [0.0873, 0.1140]	0.0884 [0.0773, 0.0995]
1024	VQC	124/88	0.1920 [0.1769, 0.2070]	0.2178 [0.2146, 0.2211]
1024	MLP	114/114	0.0776 [0.0682, 0.0870]	0.0872 [0.0858, 0.0885]

4.4 Noise and readout sensitivity

Table 10 compares each selected checkpoint pair with its same-grid, zero-noise exact-expectation control. Supplementary Table S10 merges seed identities, individual gains, offsets, and clean errors; Tables S11–S12 preserve every noise condition. These controls use the noise grid, not the E1-grid metadata stored alongside the records.

For the larger exponential-pulse circuit, mean stress error is 0.5804 ± 0.0166 clean and 2.5758 ± 0.2728, 0.9735 ± 0.0950, and 0.6299 ± 0.0251 at 10^3, 10^4, and 10^5 shots per input and observable, respectively ($n = 2$; Supplementary Table S11). At the lowest budget the larger-circuit stress multipliers span 1.642–4.438, whereas smaller-circuit multipliers span 8.637–22.396 (Table 10).

Table 10: Ratios of corrected E4 mean stress error to the corresponding same-grid clean mean. Shot columns use $p = 0$; channel columns use exact expectations. Ratios are descriptive ratios of two-seed means, not means of paired ratios or significance tests.

Pulse	$P_{\mathrm{nom}}/P_{\mathrm{out}}$	$N = 10^3$	10^4	10^5	$p = 10^{-3}$	10^{-2}
exp	44/30	22.396	7.349	2.600	0.998	0.982
exp	124/88	4.438	1.677	1.085	0.991	1.057
step	44/30	8.637	2.929	1.375	0.995	0.978
step	124/88	1.642	1.079	1.009	0.988	0.936
smooth	44/30	9.540	3.307	1.510	1.000	1.003
smooth	124/88	2.730	1.267	1.027	1.003	1.058

Exact-expectation depolarization yields larger exponential-pulse stress errors of 0.5753 ± 0.0184 at $p = 10^{-3}$ and 0.6135 ± 0.1210 at $p = 10^{-2}$ ($n = 2$; Supplementary Table S11). Across all cells the stress changes are modest (Table 10), not several-fold channel penalties. Small decreases do not establish beneficial noise. Temperature can behave differently: the smaller smooth-pulse circuit changes from 0.3249 ± 0.0189 clean to 1.2674 ± 1.3552 at $p = 10^{-2}$ ($n = 2$; Supplementary Table S11).

For a readout $z = \langle Z \rangle$ and $F = t(\alpha z + \beta)$, independent measurement statistics give

$$\mathrm{Var}(\widehat{F} \mid z) = \frac{t^2 \alpha^2 (1 - z^2)}{N}, \qquad \delta F = t\alpha\, \delta z. \tag{24}$$

Individual trained gains vary strongly even within the selected pairs (Supplementary Table S10). Large gains amplify fluctuations; offsets do not cancel them. Channel-induced bias also depends on the actual change in z and the existing approximation error, explaining why large gains alone do not imply large depolarizing penalties. Selection by best clean E1 stress error limits these findings to the evaluated checkpoints.

4.5 Secondary wave speeds

Table 11 reports the completed higher-wave-speed experiments and the same-seed subset of E1. The MLP retains lower mean temperature and stress errors than the circuit. Circuit stress error increases from 0.8274 ± 0.2146 at the original wave-speed parameter to 1.1971 ± 0.4658 and 1.4485 ± 0.8240 at the secondary values ($n = 3$; Table 11). Its largest individual error is 2.3413 (Supplementary Table S17). The growing dispersion cautions against interpreting only the means.

Every family has larger mean errors at the secondary speeds than in the full five-seed E1 table (Tables 6 and 11), but this statement changes the seed population. With the same seeds, the MLP's stress mean improves from 0.2779 to 0.2353 at the intermediate speed (Table 11). Consequently,

the evidence supports a preserved circuit/MLP ordering, not uniform degradation for every family and seed subset.

The wave front reaches the outer boundary at $t = 6/\sqrt{\gamma}$: 2, 1.2, and approximately 0.8571 for the three tested values (Tables 1 and 11). Exact temperature is independent of γ, so its changed learning errors reflect the coupled training problem. Increasing γ changes the stress normalization, initial amplitude, and relative residual scaling. In particular, Eq. (18) weights the squared unscaled stress-rate mismatch by γ^{-2}; the normalized rate target remains unchanged. Earlier front exit and changed loss scaling are relevant differences, not separately isolated causes of the observed errors. Rates were transferred without retuning, and timing shares the resource caveats of the overlapping campaign (Supplementary Tables S3 and S17).

Table 11: Secondary wave speeds: mean ± sample standard deviation of errors over seeds $0, 1, 2$ ($n = 3$). The $\gamma = 9$ rows are the same-seed subset of E1, not new runs. All runs use the exponential pulse and the E1 physics recipe; rates transfer from $\gamma = 9$: 0.1, except 0.01 for the reference. Circuit, MLP, spectral, and reference counts are 124/88, 114/114, 116/116, and 12802/12802, respectively. The reference is not parameter matched. Per-seed errors, times, and closure counts are retained in the supplement.

γ	Model	e_T	e_σ
9	VQC	0.1910 ± 0.0520	0.8274 ± 0.2146
9	MLP	0.1164 ± 0.0693	0.2779 ± 0.1511
9	Spectral	0.1895 ± 0.0683	0.6309 ± 0.2636
9	Reference	0.0117 ± 0.0038	0.0488 ± 0.0044
25	VQC	0.2927 ± 0.0458	1.1971 ± 0.4658
25	MLP	0.1279 ± 0.0525	0.2353 ± 0.1525
25	Spectral	0.3180 ± 0.0970	0.6342 ± 0.3302
25	Reference	0.0221 ± 0.0028	0.1044 ± 0.0357
49	VQC	0.3868 ± 0.1884	1.4485 ± 0.8240
49	MLP	0.2468 ± 0.0621	0.4511 ± 0.1311
49	Spectral	0.4400 ± 0.1898	0.6894 ± 0.2043
49	Reference	0.0421 ± 0.0014	0.0956 ± 0.0208

5 Discussion

5.1 What the comparisons establish

RQ1 has a negative descriptive answer under the executed recipe: the MLP has lower mean field errors in the nominal and output-affecting comparisons (Tables 6 and 7). Output-affecting matching addresses known inert coordinates, while leaving capacity and optimization differences. The result neither invalidates favorable quantum-PINN comparisons on other equations and embeddings (Berger et al., 2025) nor establishes an optimal classical or quantum architecture. Search-boundary choices and single-seed tuning further limit generalization.

RQ2 supports increased scale of one full-objective gradient statistic, without an asymptotic trainability conclusion (Table 8; Supplementary Table S9). Its fixed readouts and input derivatives differ from qubit-indexed PDE discretizations (Ukwatta Hewage et al., 2026). RQ3 has a negative answer relative to the MLP over the tested budgets: no stress-efficiency advantage appears in parameter, collocation, or production training-label sweeps conditional on separately selected

hyperparameters (Table 9). The label axis excludes the fixed tuning overhead and production validation diagnostics; it does not measure total label-acquisition efficiency (Table 3). The shallow width sequence repeats one output family with different initialization, training, and simulation cost (Supplementary Section S2). The spectral model is a weaker stress control and does not beat the circuit at every parameter target; the strongest negative comparison is with the MLP. Near equality of the deepest circuit's temperature mean does not establish parity for the coupled task. Supervised training demonstrates better attainable circuit stress accuracy, but the classical gap persists. These findings are restricted to the sampled architectures, transferred physics rates, and two seeds (Table 9; Supplementary Table S2). E1 provides the less contention-affected cost comparison; E3 timings cannot establish isolated compute scaling or project hardware speedups.

5.2 Representation, weighting, and classical alternatives

The spectral comparator tests a particular random basis, not a classical surrogate constructed from the trained circuit. It therefore cannot establish or refute dequantization (Schreiber et al., 2023; Sahebi et al., 2025). Trainable circuit encoding scales and fixed sampled frequencies differ substantially. More targeted spectral controls, learned frequencies, or surrogates of the fitted circuit would address different questions.

Wave fronts and multiple temporal scales make frequency content relevant; Fourier-feature analyses of classical PINNs provide context (Wang et al., 2021b). Unequal convergence of loss components is another documented PINN difficulty (Wang et al., 2022). Neither mechanism is diagnosed by the present field plots or aggregate losses. Adaptive weighting (Setty et al., 2025) and alternative quantum embeddings (Tran et al., 2026a) merit controlled tests, rather than attribution of the observed errors to a single presumed representational limit. The supervised E3 result strengthens this distinction: the circuit's physics-only error is not a lower bound on its representation accuracy (Table 9). It does not identify how much of the deficit arises from loss weighting, optimization, information access, or the initial-rate relaxation.

5.3 Readout amplification and limitations

Equation (24) makes unconstrained gains a practical inference concern even after exact-expectation training. Gain penalties, variance regularization, or shot-aware training could trade clean accuracy against sampling cost (Kreplin and Roth, 2024). E4 contains no intervention testing those remedies. Its fixed-checkpoint inference experiment is also distinct from noisy-gradient barren-plateau results (Wang et al., 2021a).

The study uses compact circuits and classical networks, far below many practical PINN sizes, simulated statevectors, and no hardware runs. Common iteration caps do not equalize derivative work or realized closure calls. Exact outer-boundary and validation information is available to every physics-trained family. The initial-rate strip leaves the continuum stress problem underdetermined. Its nonmonotone single-seed sensitivity differs across models: the declared cutoff gives the best tested stress outcome for the circuit and reference MLP but the worst for the matched MLP (Table 4). It is therefore neither a generally validated accuracy improvement nor a substitute for complete initial data. The dropped high-collocation circuit jobs, sparse efficiency replication, thread-sensitive reference replay, and overlapping execution conditions further limit scalability and timing conclusions (Supplementary Sections S1 and S9). The free-surface startup corner and rectangular switch-off interface also limit smooth strong-form approximation. Verification is pointwise at a single finite-difference resolution, not a uniform accuracy bound. The gradient study samples one derivative, while E4 selects clean-error minima and lacks repeated shot realizations. These

restrictions define the scope of the negative comparison.

6 Conclusions

Under the declared tuning, sampling, and initial-rate recipe, the classical MLP achieves lower mean temperature and stress errors than the tested circuit in both nominal and output-affecting count comparisons (Tables 6 and 7). The full physics objective raises a selected gradient variance, but its finite-size slopes do not establish barren-plateau mitigation (Table 8; Supplementary Table S9). The completed parameter, collocation, and production training-label sweeps show no circuit stress advantage over the MLP. The label comparison is conditional on separately selected hyperparameters and excludes tuning and diagnostic labels (Table 3). Supervision reveals better attainable circuit accuracy than the physics recipe achieved (Table 9). Secondary speeds preserve the circuit/MLP ordering, with larger circuit stress dispersion (Table 11). Trained readout gains produce substantial finite-shot sensitivity in selected circuits, whereas the tested exact-expectation depolarization changes their stress errors modestly (Table 10).

For laser processing, thermal-stress control, and thermal barrier coating assessment, the practical lesson is to verify stress initial data, normalization, field accuracy, and computational cost together. The corrected reference fields and explicit conversion to pascals support that verification. Extension to realistic materials requires complete initial-rate treatment and richer thermal-mechanical physics; this benchmark establishes neither quantum advantage nor a calibrated engineering design tool.

Data and code availability

The assembled archive `QPINN_LTE_release_v1_2026-10-21.zip` contains the analysis and reference code, all launchers and generators, parameter-audit and family-equivalence scripts, and the four verification scripts. Its `results/` directory contains the 440 retained per-run JSON records collected in `runs_all.json`, each with its checkpoint, and 24 further JSON files of other types: 10 superseded per-run records from discarded launches (segregated, used by no table), the current and superseded E2/E4 aggregate files, three selected-rate summaries, and seven pilot and diagnostic files (464 JSON files in all). It also includes 456 checkpoint state-dictionary files (`.pt`, 6.9 MB of content: 440 retained, 10 superseded, 6 diagnostic), the E2 aggregate records (`e2_exp_g9.json`), and the corrected checkpoint-labeled E4 records (`e4_noise.json`). Execution logs in `logs/` (527 files, 0.45 MB) include queue dispatchers and the initial-rate weight-ablation logs and variants. The package contains results package v6, tables and figures, histories and closure counts, verification and audit outputs, the family-equivalence check, checkpoint gains, and selected rates (Supplementary Section S1 and Table S1). The `RELEASE_README` maps every manuscript table and figure to its inputs and generator and records regeneration commands and launcher thread settings; Supplementary Sections S8–S9 give reconstruction details and individual E3 and secondary-speed outcomes. Repository: `https://github.com/Alqahtani921893/qpinn-laser-thermoelasticity-release` (release v1, 21 October 2026); the archived release receives a Zenodo DOI at acceptance.

CRediT

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Declaration of competing interests

The author declares no competing interests.

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A Corrected closed-form solution and verification

The expressions below define the exact reference; Supplementary Section S7 gives the correction inventory and executed pointwise checks. The transformed initial-rate terms must be retained before cancellation.

A.1 Transform solution and the initial stress rate

Let $\widehat{f}(s) = \mathcal{L}[f](s)$, $a = \sqrt{\gamma}$, and $\tau = x/a$, using the branch of $\sqrt{s}$ positive on the positive real axis. Zero initial fields and $T_t(x, 0^+) = e^{-x} f(0^+)$ together with $\sigma_t(x, 0^+) = -\gamma e^{-x} f(0^+)$ give

$$\widehat{\sigma}_{xx} - \frac{s^2}{\gamma}\widehat{\sigma} = s^2 \widehat{T}. \tag{25}$$

The initial-rate terms on the two sides cancel. Solving the spatial boundary value problems gives

$$\widehat{T}(x, s) = \widehat{f}(s) \left[\frac{e^{-x}}{s-1} - \frac{e^{-x\sqrt{s}}}{\sqrt{s}(s-1)} \right], \tag{26}$$

$$\widehat{\sigma}(x, s) = \widehat{f}(s) \left[\mathcal{A}(s)\left(e^{-x} - e^{-s\tau}\right) - \mathcal{B}(s)\left(e^{-x\sqrt{s}} - e^{-s\tau}\right) \right], \tag{27}$$

$$\mathcal{A}(s) = \frac{\gamma s^2}{(s-1)(\gamma - s^2)}, \qquad \mathcal{B}(s) = \frac{\gamma\sqrt{s}}{(s-1)(\gamma - s)}. \tag{28}$$

The apparent positive poles at $s = 1$, $s = a$, and $s = \gamma$ cancel in the complete expression, so they do not produce growing physical modes. At $s = 1$ the cancellation is between terms; at $s = a$ and $s = \gamma$ the corresponding exponential differences vanish. The analogous assertion that the first difference vanishes at $s = -a$ would be incorrect.

A.2 Exponential response and pulse superposition

Let $\mathcal{T}_\beta$ and $\mathcal{S}_\beta$ denote the responses to $e^{-\beta t} H(t)$, with unit amplitude. For $t > 0$, define

$$\xi = \frac{x}{2\sqrt{t}}, \qquad \eta = \sqrt{\beta t}, \tag{29}$$

$$J_\beta(x, t) = e^{-\beta t} \operatorname{Im}\left[e^{i\sqrt{\beta} x} \operatorname{erfc}(\xi + i\eta) \right], \tag{30}$$

$$C_1 = \frac{\gamma}{(\gamma - 1)(1 + \beta)}, \qquad C_2 = \frac{\gamma}{(\gamma - 1)(\gamma + \beta)}, \qquad C_3 = -\frac{\gamma}{(1 + \beta)(\gamma + \beta)}, \tag{31}$$

$$r_+ = -\frac{\gamma a}{2(a - 1)(a + \beta)}, \qquad r_- = \frac{\gamma a}{2(a + 1)(a - \beta)}, \qquad r_\beta = \frac{\gamma \beta^2}{(1 + \beta)(\beta^2 - \gamma)}. \tag{32}$$

For $\beta > 0$,

$$\mathcal{T}_\beta(x,t) = \frac{1}{1+\beta}\left[\frac{e^{t-x}}{2}\operatorname{erfc}(\sqrt{t}-\xi) + \frac{e^{t+x}}{2}\operatorname{erfc}(\sqrt{t}+\xi) - e^{-x-\beta t} - \frac{J_\beta(x,t)}{\sqrt{\beta}}\right]. \tag{33}$$

The stress can be written without the canceling Gaussian terms, since $C_1 - C_2 + C_3 = 0$. Define

$$P_1(x,t) = \frac{1}{2}\left[e^{t+x}\operatorname{erfc}(\xi+\sqrt{t}) + e^{t-x}\operatorname{erfc}(\sqrt{t}-\xi)\right], \tag{34}$$

$$P_a(x,t) = \frac{a}{2}\left[e^{\gamma t-ax}\operatorname{erfc}(\xi-a\sqrt{t}) - e^{\gamma t+ax}\operatorname{erfc}(\xi+a\sqrt{t})\right], \tag{35}$$

$$R_\beta(t) = r_\beta e^{-\beta t} + r_+ e^{at} + r_- e^{-at}, \tag{36}$$

$$D_\beta(z) = C_1 e^z \operatorname{erfc}(\sqrt{z}) + aC_2 e^{\gamma z}\operatorname{erf}(a\sqrt{z}) + C_3\sqrt{\beta}e^{-\beta z}\operatorname{erfi}(\sqrt{\beta z}) + R_\beta(z). \tag{37}$$

Then

$$\mathcal{S}_\beta(x,t) = C_1 P_1(x,t) + C_2 P_a(x,t) - C_3\sqrt{\beta}J_\beta(x,t) + e^{-x}R_\beta(t) - H(t-\tau)D_\beta(t-\tau). \tag{38}$$

The delayed term is evaluated only for $t > \tau$ and extended continuously at the front. The formulas are interpreted by limits at coincident poles, including $\gamma = 1$ and $\gamma = \beta^2$. The supplied numerical implementation does not implement these general limiting cases; the benchmark parameters in Table 1 avoid them. The unit-step stress is $\mathcal{S}_0 = \lim_{\beta\to 0^+}\mathcal{S}_\beta$. For the temperature this limit is explicitly

$$\begin{aligned}\mathcal{T}_0(x,t) = &-e^{-x} + \frac{e^{t-x}}{2}\operatorname{erfc}(\sqrt{t}-\xi) + \frac{e^{t+x}}{2}\operatorname{erfc}(\sqrt{t}+\xi)\\ &+ 2\sqrt{\frac{t}{\pi}}e^{-\xi^2} - x\operatorname{erfc}(\xi).\end{aligned} \tag{39}$$

All responses vanish for nonpositive time. For either field $\mathcal{F} \in \{\mathcal{T}, \mathcal{S}\}$, the benchmark pulses follow by linearity:

$$F_{\exp}(x,t) = A_e\mathcal{F}_\beta(x,t), \tag{40}$$

$$F_{\text{step}}(x,t) = A_s\left[\mathcal{F}_0(x,t) - H(t-\Delta)\mathcal{F}_0(x,t-\Delta)\right], \tag{41}$$

$$F_{\text{smooth}}(x,t) = A\left[\mathcal{F}_{\beta_1}(x,t) - \mathcal{F}_{\beta_2}(x,t)\right]. \tag{42}$$

For pulses nonzero at startup, the delayed stress response has a derivative discontinuity at $x = at$. The rectangular pulse also launches a delayed front from its switch-off time. The smooth pulse cancels the leading startup discontinuity because $f(0^+) = 0$.

For numerical evaluation, the identity $e^{c^2t\pm cx}\operatorname{erfc}(\xi \pm c\sqrt{t}) = e^{-\xi^2}\operatorname{erfcx}(\xi \pm c\sqrt{t})$ is useful, but negative large arguments and cancellation between delayed terms still require analytic regrouping. In particular, replacing every factor individually by a scaled complementary error function does not by itself remove all large intermediate terms. The released reference implementation regroups the canceling $e^{\gamma t}$ contributions behind the front.

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